HR_dataset

March 20, 2026

[1]:
```
pip install pandas numpy matplotlib seaborn
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pandas in
c:\users\annap\appdata\roaming\python\python313\site-packages (2.3.2)
Requirement already satisfied: numpy in
c:\users\annap\appdata\roaming\python\python313\site-packages (2.3.3)
Requirement already satisfied: matplotlib in
c:\users\annap\appdata\roaming\python\python313\site-packages (3.10.6)
Requirement already satisfied: seaborn in
c:\users\annap\appdata\roaming\python\python313\site-packages (0.13.2)
Requirement already satisfied: python-dateutil>=2.8.2 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from pandas)
(2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from pandas)
(2025.2)
Requirement already satisfied: tzdata>=2022.7 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from pandas)
(2025.2)
Requirement already satisfied: contourpy>=1.0.1 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from matplotlib)
(1.3.3)
Requirement already satisfied: cycler>=0.10 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from matplotlib)
(0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from matplotlib)
(4.59.2)
Requirement already satisfied: kiwisolver>=1.3.1 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from matplotlib)
(1.4.9)
Requirement already satisfied: packaging>=20.0 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from matplotlib)
(25.0)
Requirement already satisfied: pillow>=8 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from matplotlib)
(11.3.0)
```

```
Requirement already satisfied: pyparsing>=2.3.1 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from matplotlib)
(3.2.4)
Requirement already satisfied: six>=1.5 in
c:\users\annap\appdata\roaming\python\python313\site-packages (from python-
dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.


[notice] A new release of pip is available: 25.2 -> 26.0.1
[notice] To update, run: C:\Program Files\Python313\python.exe -m pip install
--upgrade pip
```

[4]:
```python
import pandas as pd

df = pd.read_csv(r"C:\Users\annap\OneDrive\Desktop\Excel\HR_Data.csv")

df.head()
```

[4]:
```
   EmpID  Age AgeGroup Attrition     BusinessTravel  DailyRate  \
0  RM297   18    18-25       Yes      Travel_Rarely        230
1  RM302   18    18-25        No      Travel_Rarely        812
2  RM458   18    18-25       Yes  Travel_Frequently       1306
3  RM728   18    18-25        No         Non-Travel        287
4  RM829   18    18-25       Yes         Non-Travel        247

               Department  DistanceFromHome  Education EducationField  …  \
0  Research & Development                 3          3  Life Sciences  …
1                   Sales                10          3        Medical  …
2                   Sales                 5          3      Marketing  …
3  Research & Development                 5          2  Life Sciences  …
4  Research & Development                 8          1        Medical  …

   RelationshipSatisfaction  StandardHours  StockOptionLevel  \
0                         3             80                 0
1                         1             80                 0
2                         4             80                 0
3                         4             80                 0
4                         4             80                 0

  TotalWorkingYears  TrainingTimesLastYear  WorkLifeBalance  YearsAtCompany  \
0                 0                      2                3               0
1                 0                      2                3               0
2                 0                      3                3               0
3                 0                      2                3               0
4                 0                      0                3               0

  YearsInCurrentRole  YearsSinceLastPromotion YearsWithCurrManager
```

```
0                   0                         0                   0.0
1                   0                         0                   0.0
2                   0                         0                   0.0
3                   0                         0                   0.0
4                   0                         0                   0.0

[5 rows x 38 columns]
```

[4]:
```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1480 entries, 0 to 1479
Data columns (total 38 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   EmpID                     1480 non-null   object
 1   Age                       1480 non-null   int64
 2   AgeGroup                  1480 non-null   object
 3   Attrition                 1480 non-null   object
 4   BusinessTravel            1480 non-null   object
 5   DailyRate                 1480 non-null   int64
 6   Department                1480 non-null   object
 7   DistanceFromHome          1480 non-null   int64
 8   Education                 1480 non-null   int64
 9   EducationField            1480 non-null   object
 10  EmployeeCount             1480 non-null   int64
 11  EmployeeNumber            1480 non-null   int64
 12  EnvironmentSatisfaction   1480 non-null   int64
 13  Gender                    1480 non-null   object
 14  HourlyRate                1480 non-null   int64
 15  JobInvolvement            1480 non-null   int64
 16  JobLevel                  1480 non-null   int64
 17  JobRole                   1480 non-null   object
 18  JobSatisfaction           1480 non-null   int64
 19  MaritalStatus             1480 non-null   object
 20  MonthlyIncome             1480 non-null   int64
 21  SalarySlab                1480 non-null   object
 22  MonthlyRate               1480 non-null   int64
 23  NumCompaniesWorked        1480 non-null   int64
 24  Over18                    1480 non-null   object
 25  OverTime                  1480 non-null   object
 26  PercentSalaryHike         1480 non-null   int64
 27  PerformanceRating         1480 non-null   int64
 28  RelationshipSatisfaction  1480 non-null   int64
 29  StandardHours             1480 non-null   int64
 30  StockOptionLevel          1480 non-null   int64
 31  TotalWorkingYears         1480 non-null   int64
 32  TrainingTimesLastYear     1480 non-null   int64
```

```
 33  WorkLifeBalance           1480 non-null   int64
 34  YearsAtCompany            1480 non-null   int64
 35  YearsInCurrentRole        1480 non-null   int64
 36  YearsSinceLastPromotion   1480 non-null   int64
 37  YearsWithCurrManager      1423 non-null   float64
dtypes: float64(1), int64(25), object(12)
memory usage: 439.5+ KB
```

[5]:
```
df.isnull().sum()
```

[5]:
```
EmpID                       0
Age                         0
AgeGroup                    0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
SalarySlab                  0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
```

```
YearsWithCurrManager        57
dtype: int64
```

[6]:
```
df.head()
```

[6]:
```
   EmpID  Age AgeGroup Attrition     BusinessTravel  DailyRate  \
0  RM297   18    18-25       Yes      Travel_Rarely        230
1  RM302   18    18-25        No      Travel_Rarely        812
2  RM458   18    18-25       Yes  Travel_Frequently       1306
3  RM728   18    18-25        No         Non-Travel        287
4  RM829   18    18-25       Yes         Non-Travel        247

               Department  DistanceFromHome  Education EducationField  …  \
0  Research & Development                 3          3  Life Sciences  …
1                   Sales                10          3        Medical  …
2                   Sales                 5          3      Marketing  …
3  Research & Development                 5          2  Life Sciences  …
4  Research & Development                 8          1        Medical  …

   RelationshipSatisfaction  StandardHours  StockOptionLevel  \
0                         3             80                 0
1                         1             80                 0
2                         4             80                 0
3                         4             80                 0
4                         4             80                 0

  TotalWorkingYears  TrainingTimesLastYear  WorkLifeBalance  YearsAtCompany  \
0                 0                      2                3               0
1                 0                      2                3               0
2                 0                      3                3               0
3                 0                      2                3               0
4                 0                      0                3               0

  YearsInCurrentRole  YearsSinceLastPromotion YearsWithCurrManager
0                  0                        0                  0.0
1                  0                        0                  0.0
2                  0                        0                  0.0
3                  0                        0                  0.0
4                  0                        0                  0.0

[5 rows x 38 columns]
```

[6]:
```
df.shape
```

[6]:
```
(1480, 38)
```

[8]:
```
df = df.drop(columns=[
'EmpID',
'EmployeeNumber',
'EmployeeCount',
'Over18',
'StandardHours'
])
```

[9]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1480 entries, 0 to 1479
Data columns (total 33 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Age                       1480 non-null   int64
 1   AgeGroup                  1480 non-null   object
 2   Attrition                 1480 non-null   object
 3   BusinessTravel            1480 non-null   object
 4   DailyRate                 1480 non-null   int64
 5   Department                1480 non-null   object
 6   DistanceFromHome          1480 non-null   int64
 7   Education                 1480 non-null   int64
 8   EducationField            1480 non-null   object
 9   EnvironmentSatisfaction   1480 non-null   int64
 10  Gender                    1480 non-null   object
 11  HourlyRate                1480 non-null   int64
 12  JobInvolvement            1480 non-null   int64
 13  JobLevel                  1480 non-null   int64
 14  JobRole                   1480 non-null   object
 15  JobSatisfaction           1480 non-null   int64
 16  MaritalStatus             1480 non-null   object
 17  MonthlyIncome             1480 non-null   int64
 18  SalarySlab                1480 non-null   object
 19  MonthlyRate               1480 non-null   int64
 20  NumCompaniesWorked        1480 non-null   int64
 21  OverTime                  1480 non-null   object
 22  PercentSalaryHike         1480 non-null   int64
 23  PerformanceRating         1480 non-null   int64
 24  RelationshipSatisfaction  1480 non-null   int64
 25  StockOptionLevel          1480 non-null   int64
 26  TotalWorkingYears         1480 non-null   int64
 27  TrainingTimesLastYear     1480 non-null   int64
 28  WorkLifeBalance           1480 non-null   int64
 29  YearsAtCompany            1480 non-null   int64
 30  YearsInCurrentRole        1480 non-null   int64
 31  YearsSinceLastPromotion   1480 non-null   int64
 32  YearsWithCurrManager      1423 non-null   float64
```

```
dtypes: float64(1), int64(22), object(10)
memory usage: 381.7+ KB
```

[7]:
```python
df.isnull().sum()
```

[7]:
```
EmpID                       0
Age                         0
AgeGroup                    0
Attrition                   0
BusinessTravel              0
DailyRate                   0
Department                  0
DistanceFromHome            0
Education                   0
EducationField              0
EmployeeCount               0
EmployeeNumber              0
EnvironmentSatisfaction     0
Gender                      0
HourlyRate                  0
JobInvolvement              0
JobLevel                    0
JobRole                     0
JobSatisfaction             0
MaritalStatus               0
MonthlyIncome               0
SalarySlab                  0
MonthlyRate                 0
NumCompaniesWorked          0
Over18                      0
OverTime                    0
PercentSalaryHike           0
PerformanceRating           0
RelationshipSatisfaction    0
StandardHours               0
StockOptionLevel            0
TotalWorkingYears           0
TrainingTimesLastYear       0
WorkLifeBalance             0
YearsAtCompany              0
YearsInCurrentRole          0
YearsSinceLastPromotion     0
YearsWithCurrManager       57
dtype: int64
```

[8]:
```python
#Convert Attrition to Numeric
df["Attrition"] = df["Attrition"].map({"Yes":1,"No":0})
```

[13]:
```
df.head()
```

[13]:
```
   Age AgeGroup  Attrition     BusinessTravel  DailyRate  \
0   18    18-25          1      Travel_Rarely        230
1   18    18-25          0      Travel_Rarely        812
2   18    18-25          1  Travel_Frequently       1306
3   18    18-25          0         Non-Travel        287
4   18    18-25          1         Non-Travel        247

                Department  DistanceFromHome  Education EducationField  \
0  Research & Development                 3          3  Life Sciences
1                   Sales                10          3        Medical
2                   Sales                 5          3      Marketing
3  Research & Development                 5          2  Life Sciences
4  Research & Development                 8          1        Medical

   EnvironmentSatisfaction  … PerformanceRating  RelationshipSatisfaction  \
0                        3  …                 3                         3
1                        4  …                 3                         1
2                        2  …                 3                         4
3                        2  …                 3                         4
4                        3  …                 3                         4

   StockOptionLevel  TotalWorkingYears TrainingTimesLastYear  WorkLifeBalance  \
0                 0                  0                     2                3
1                 0                  0                     2                3
2                 0                  0                     3                3
3                 0                  0                     2                3
4                 0                  0                     0                3

  YearsAtCompany  YearsInCurrentRole YearsSinceLastPromotion  \
0              0                   0                        0
1              0                   0                        0
2              0                   0                        0
3              0                   0                        0
4              0                   0                        0

   YearsWithCurrManager
0                   0.0
1                   0.0
2                   0.0
3                   0.0
4                   0.0

[5 rows x 33 columns]
```

[9]:
```
#What is the overall attrition rate?
attrition_rate = df["Attrition"].mean()*100
print("Attrition Rate:", attrition_rate)
```

```
Attrition Rate: 16.08108108108108
```

[10]:
```
#Which department has highest attrition?
df.groupby("Department")["Attrition"].mean().sort_values(ascending=False)
```

[10]:
```
Department
Sales                      0.206667
Human Resources            0.190476
Research & Development     0.137539
Name: Attrition, dtype: float64
```

[11]:
```
#Does overtime increase attrition?
df.groupby("OverTime")["Attrition"].mean()
```

[11]:
```
OverTime
No      0.103578
Yes     0.306220
Name: Attrition, dtype: float64
```

[12]:
```
#Which age group has highest attrition?
df.groupby("AgeGroup")["Attrition"].mean().sort_values(ascending=False)
```

[12]:
```
AgeGroup
18-25    0.357724
26-35    0.189853
55+      0.170213
46-55    0.118421
36-45    0.091295
Name: Attrition, dtype: float64
```

[14]:
```
#Average salary by department
df.groupby("Department")["MonthlyIncome"].mean()
```

[14]:
```
Department
Human Resources            6654.507937
Research & Development     6280.365047
Sales                      6966.737778
Name: MonthlyIncome, dtype: float64
```

[15]:
```
#Attrition by job role
df.groupby("JobRole")["Attrition"].mean().sort_values(ascending=False)
```

[15]:
```
JobRole
Sales Representative         0.392857
```

```
Laboratory Technician        0.237548
Human Resources              0.230769
Sales Executive              0.176292
Research Scientist           0.160410
Healthcare Representative    0.068182
Manufacturing Director       0.068027
Manager                      0.049020
Research Director            0.025000
Name: Attrition, dtype: float64
```

[16]:
```
#Employees working overtime
df["OverTime"].value_counts()
```

[16]:
```
OverTime
No     1062
Yes     418
Name: count, dtype: int64
```

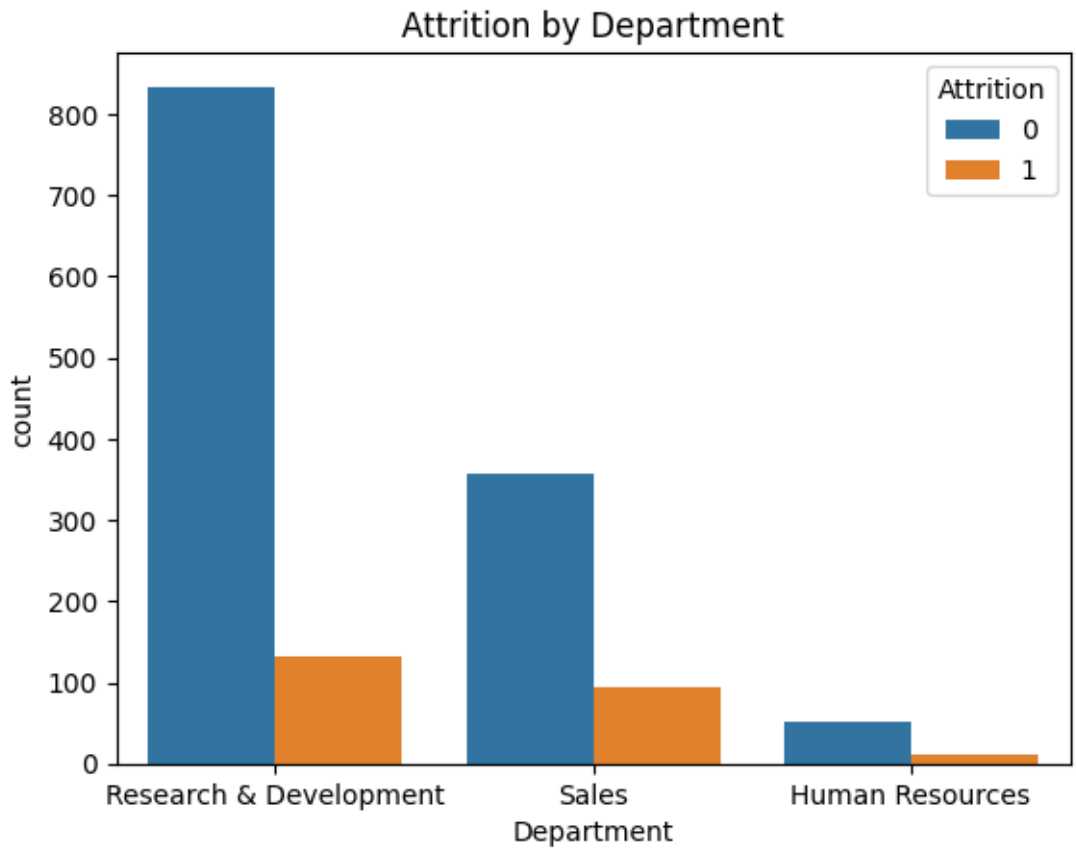
[17]:
```
#Relationship between salary and attrition
df.groupby("SalarySlab")["Attrition"].mean()
```

[17]:
```
SalarySlab
10k-15k    0.140000
15k+       0.037594
5k-10k     0.110360
Upto 5k    0.216467
Name: Attrition, dtype: float64
```

[18]:
```
#Attrition by department in seaborn
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(x="Department", hue="Attrition", data=df)

plt.title("Attrition by Department")
plt.show()
```



[19]:
```
#Correlation Analysis
corr = df.corr(numeric_only=True)

sns.heatmap(corr, annot=True)
plt.show()
```

[20]:
```
#Attrition Rate
df["Attrition"].value_counts()
```

[20]:
```
Attrition
0    1242
1     238
Name: count, dtype: int64
```

[21]:
```
#Average Salary by Job Role
df.groupby("JobRole")["MonthlyIncome"].mean()
```

[21]:
```
JobRole
Healthcare Representative     7547.416667
Human Resources               4235.750000
Laboratory Technician         3239.088123
Manager                      17181.676471
Manufacturing Director        7305.258503
```

```
Research Director            16033.550000
Research Scientist            3241.832765
Sales Executive               6947.082067
Sales Representative          2630.047619
Name: MonthlyIncome, dtype: float64
```

[22]:
```python
#Average Experience by Department
df.groupby("Department")["TotalWorkingYears"].mean()
```

[22]:
```
Department
Human Resources           11.555556
Research & Development    11.326784
Sales                     11.146667
Name: TotalWorkingYears, dtype: float64
```

[23]:
```python
#Job satisfaction vs attrition
df.groupby("JobSatisfaction")["Attrition"].value_counts()
```

[23]:
```
JobSatisfaction  Attrition
1                0            226
                 1             67
2                0            236
                 1             46
3                0            371
                 1             73
4                0            409
                 1             52
Name: count, dtype: int64
```

[24]:
```python
#Distance vs attrition
df.groupby("DistanceFromHome")["Attrition"].value_counts()
```

[24]:
```
DistanceFromHome  Attrition
1                 0            182
                  1             26
2                 0            184
                  1             28
3                 0             70
                  1             14
4                 0             56
                  1              9
5                 0             57
                  1             10
6                 0             53
                  1              7
7                 0             73
                  1             11
```

```
8                   0              71
                    1              10
9                   0              67
                    1              18
10                  0              75
                    1              11
11                  0              25
                    1               4
12                  0              14
                    1               6
13                  0              13
                    1               6
14                  0              17
                    1               4
15                  0              21
                    1               5
16                  0              25
                    1               7
17                  0              15
                    1               5
18                  0              22
                    1               4
19                  0              19
                    1               3
20                  0              21
                    1               4
21                  0              15
                    1               3
22                  0              13
                    1               6
23                  0              23
                    1               5
24                  0              17
                    1              12
25                  0              19
                    1               6
26                  0              22
                    1               3
27                  0               9
                    1               3
28                  0              22
                    1               3
29                  0              22
                    1               5
Name: count, dtype: int64
```

[25]:
```
#Promotion vs attrition
df.groupby("YearsSinceLastPromotion")["Attrition"].value_counts()
```

[25]:
```
YearsSinceLastPromotion  Attrition
0                        0            476
                         1            110
1                        0            311
                         1             49
2                        0            132
                         1             28
3                        0             43
                         1              9
4                        0             56
                         1              5
5                        0             43
                         1              2
6                        0             26
                         1              6
7                        0             60
                         1             16
8                        0             18
9                        0             14
                         1              4
10                       0              5
                         1              1
11                       0             22
                         1              2
12                       0             10
13                       0              8
                         1              2
14                       0              8
                         1              1
15                       0             10
                         1              3
Name: count, dtype: int64
```

[ ]:
```

```